

Supplementary Material

Three Equivalent Routes to Estimating Power Spectra: Averaging, Multitaper, and Smoothing

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Figure, table and equation numbers without an “S” refer to the main text, whose notation (Section 2.1 there) is used throughout: R is the design resolution, $\alpha = N\Delta R/2$ the time–bandwidth product, L the number of tapers, and widths are in units of $1/(N\Delta)$.

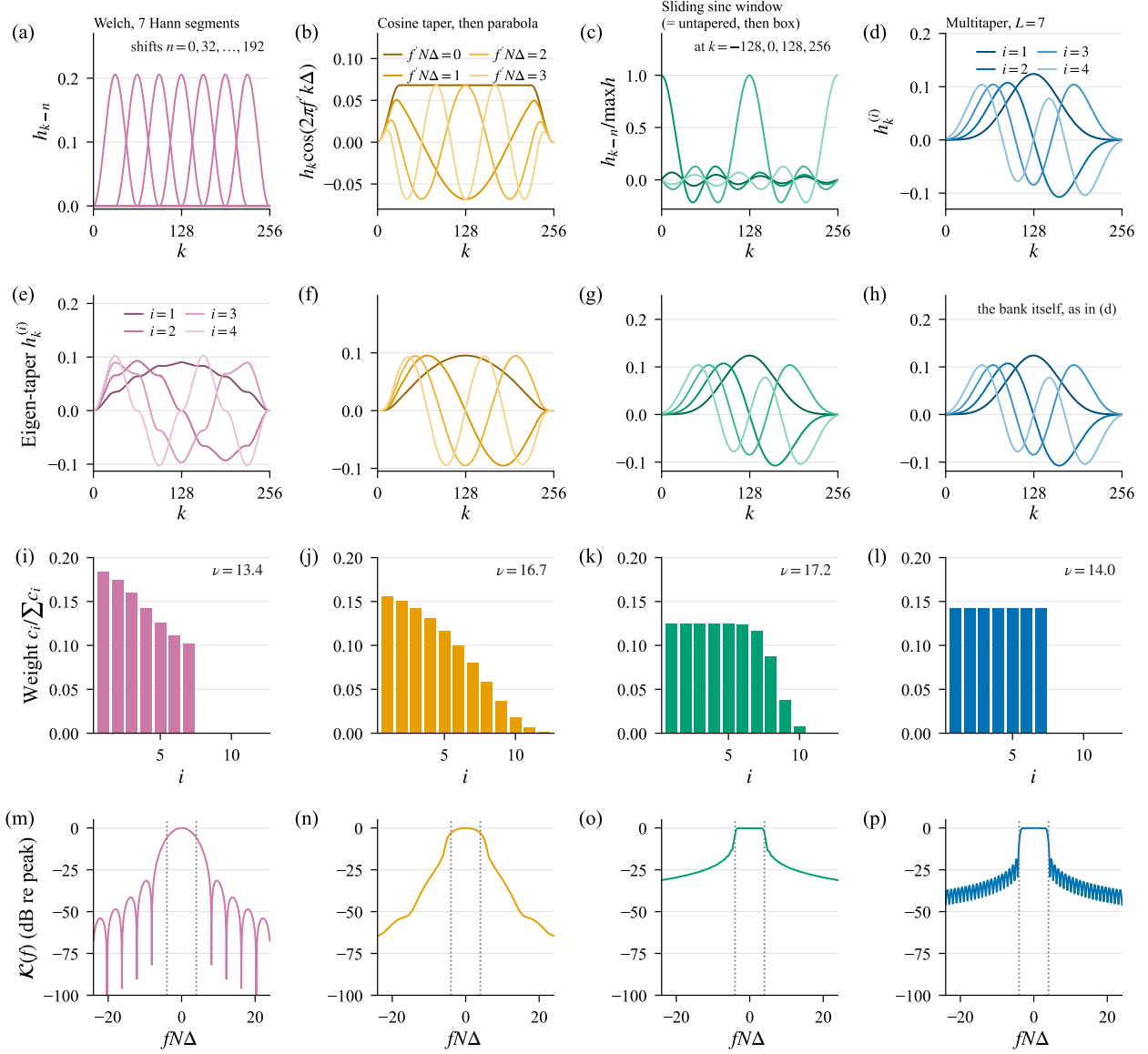


Figure S1. Fig. 5 in full. Columns and colours as in Fig. 5; rows: the bank as constructed (a–d), the first four eigen-tapers of Q (e–h; for the sliding sinc they are the Slepian sequences, and for Thomson’s estimate, whose weights are equal, the Slepian sequences are one of many orthonormal bases of the span), the eigen-weight ladder (i–l), and the kernel in dB relative to its peak (m–p; dotted lines at $|f| = R/2$). The half-power widths of the kernels are 5.84, 8.03, 8.03 and 7.41/($N\Delta$), and 0.05%, 0.13%, 1.39% and 0.22% of their power lies outside $|f| \leq R$.

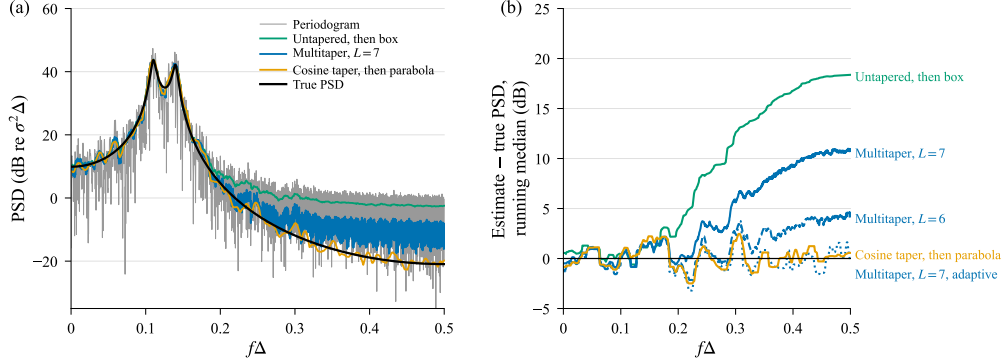


Figure S2. Dropping the leakiest taper. (a) One realization of the AR(4) process ($N = 1024$, $R = 8/(N\Delta)$): the periodogram ($\nu = 2$, grey); the untapered periodogram smoothed with the box ($\nu = 17.2$, green); the equal-weight multitaper estimate with $L = 7$ ($\nu = 14$, blue); the cosine-tapered periodogram smoothed with a parabola of half-power width R ($\nu = 16.7$, orange); and the true PSD (black). Spectra are in dB relative to $\sigma^2\Delta$, where σ^2 is the variance of the innovations. (b) Each smoothed estimate minus the true PSD, as a running median over $0.02/\Delta$ so that the levels can be read, named at its right-hand end. Panel (b) adds the equal-weight multitaper estimate with $L = 6$ ($\nu = 12$, blue, dashed) and the adaptively weighted estimate with $L = 7$ (blue, dotted; its ν varies with frequency, from 14 where the spectrum is high to 6.5 where it is low), which cannot be told apart from their neighbours in (a). For $f\Delta \geq 0.4$ the untapered box lies 18 dB above the true PSD in the median and the multitaper estimate with $L = 7$ lies 10 dB above it. Dropping the seventh taper lowers that to 4 dB, and the adaptive weights and the smoothed periodogram track the truth.

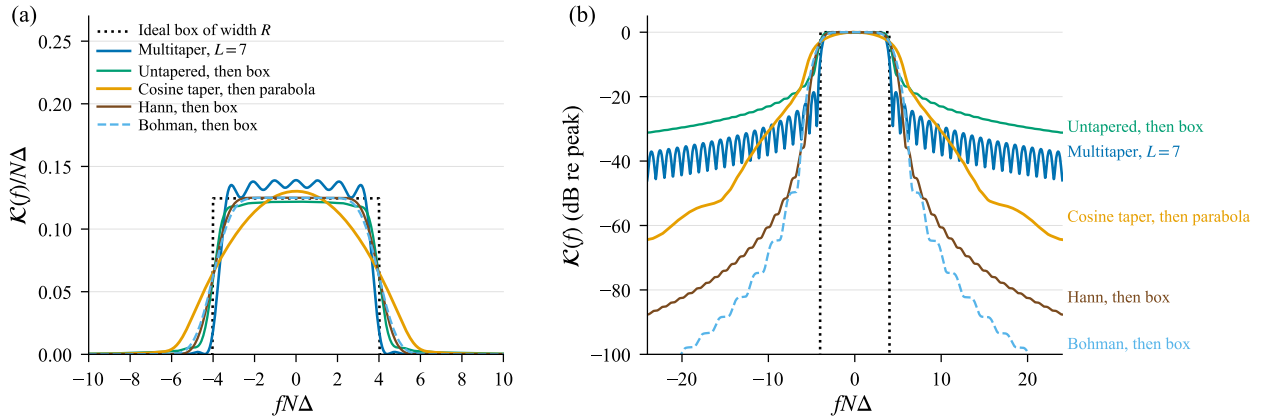


Figure S3. Kernels at the same design resolution ($N = 256$, $R = 8/(N\Delta)$; Table 2), computed exactly, (a) on a linear scale near the main lobe and (b) in dB relative to the peak: the ideal box (dotted); the $L = 7$ multitaper kernel ($\nu = 14.0$, blue); the periodogram after a 25% cosine taper followed by a parabola of the same half-power width ($\nu = 16.7$, orange), which is recipe (b'); and the untapered ($\nu = 17.2$, green), Hann ($\nu = 8.9$, brown) and Bohman ($\nu = 7.3$, light blue, dashed) periodograms followed by the box. In (b) each curve is named at its right-hand end. The half-power widths are $7.4/(N\Delta)$ for the multitaper kernel and $8.0/(N\Delta)$ for the others. At $|f| = 12/(N\Delta)$ the skirts lie between -24 dB (untapered) and -75 dB (Bohman), a range of 51 dB.

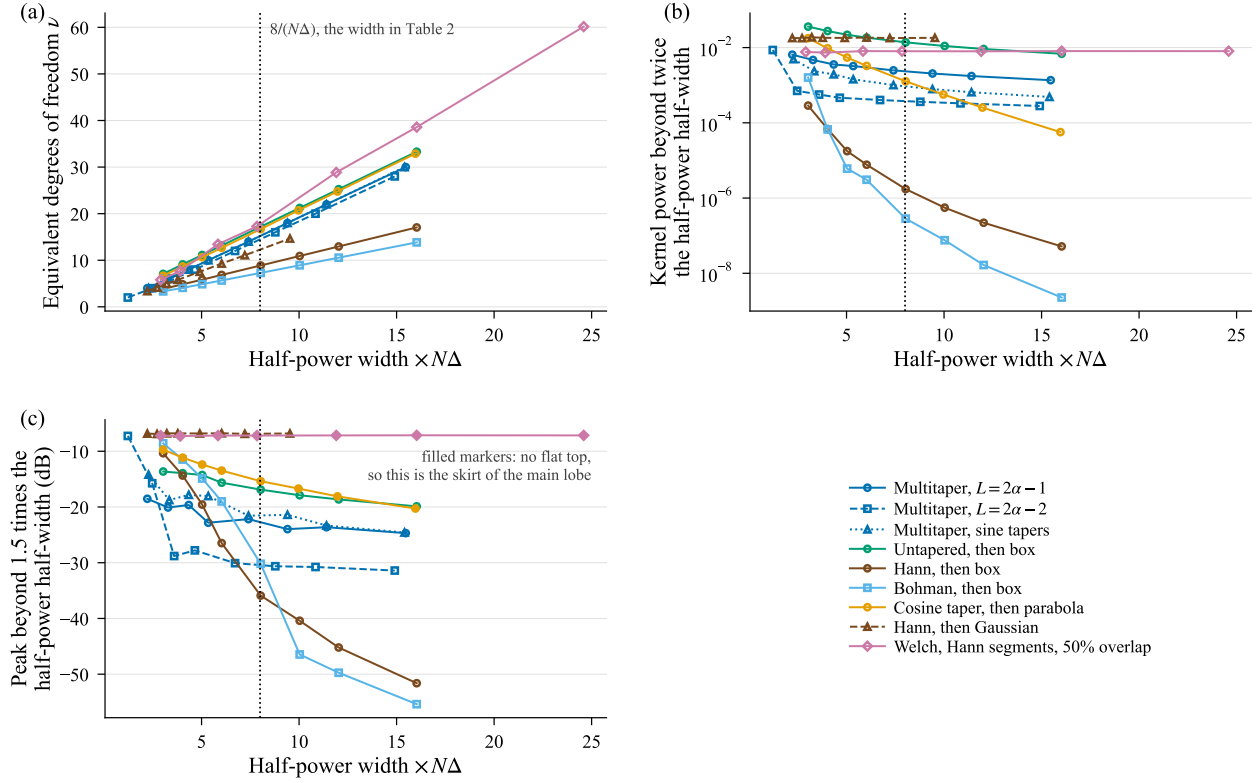


Figure S4. Variance, leakage and side-lobe level against the measured resolution, for $N = 256$. All curves are exact kernel and degrees-of-freedom computations, with no sampling variability. Each method's bandwidth parameter is swept and the quantities are plotted against the resulting half-power width, so methods are compared at equal resolution: multitaper with $L = 2\alpha - 1$ and $2\alpha - 2$ Slepian tapers and with sine tapers; untapered, Hann and Bohman periodograms followed by a box; a cosine-tapered periodogram followed by a parabola; a Hann periodogram followed by a Gaussian; Welch's method with 50% overlap. (a) Equivalent degrees of freedom ν (higher means less variance). (b) Kernel power beyond twice the half-power half-width. (c) The peak of the kernel beyond $1.5\times$ the half-power half-width, in dB relative to its maximum. The Gaussian, parabolic and Welch kernels have no flat top, so for them this quantity is the skirt of the main lobe and not a side lobe, and their markers are filled. The dotted line marks the $8/(N\Delta)$ width of Table 2, and the legend is at the lower right.